

Activity 2: AI-Assisted Unit Test Design and Mock Data Generation

Purpose

Learn how to use AI to design a testing strategy, generate comprehensive test cases, prepare realistic mock data, and create unit tests for an existing application. Developers will also learn to validate AI-generated test artifacts before using them.

Pre-requisites

- Completion of Activity 1
- Refactored EMI Calculator application
- VS Code with Cline configured

Scenario

The SmartBank EMI Calculator has been successfully developed and refactored. Before the application is released, the QA Lead has requested comprehensive test coverage to ensure the business logic is reliable and all user scenarios have been considered.

As a developer, your responsibility is to use AI to prepare the testing artifacts that will assist the QA team during validation.

Note: The objective of this activity is to learn how AI assists in designing and generating test artifacts. Configuring and executing a JavaScript testing framework is outside the scope of this activity.

Tasks

Task 1: Prepare a Testing Strategy

Analyze the EMI Calculator and identify everything that should be tested before creating test cases.

Suggested Prompt

Act as a Senior QA Engineer reviewing the Oracle JET EMI Calculator application.

Analyze the application and prepare a testing strategy.

Include:

- Features that should be tested
- Business rules
- Input validation requirements

- Positive scenarios
- Negative scenarios
- Boundary conditions
- Error handling scenarios
- Potential risks

Explain your findings in a structured format.

Do not generate test cases or code.

Task 2: Generate Comprehensive Test Cases

Create a structured set of test cases based on the testing strategy.

Suggested Prompt

Using the testing strategy prepared earlier, generate a comprehensive set of functional test cases.

For each test case include:

- Test Case ID
- Feature
- Scenario
- Test Steps
- Test Data
- Expected Result
- Priority (High, Medium, Low)
- Test Type (Positive, Negative, Boundary)

Generate at least 15 meaningful test cases.

Present the output as a table.

Do not generate unit test code.

Task 3: Generate Mock Test Data

Prepare realistic business data that can be used to execute the generated test cases.

Suggested Prompt

Generate realistic mock data for testing the EMI Calculator.

Include multiple combinations of:

- Loan Amount
- Annual Interest Rate
- Loan Tenure

Cover different loan categories such as:

- Personal Loan
- Home Loan
- Car Loan
- Education Loan

Include both valid and invalid input values.

Present the output as a well-structured table.

Task 4: Generate Unit Tests

Generate unit tests for the EMI calculation logic using AI.

Suggested Prompt

Review the EMI Calculator implementation.

Generate unit tests for the EMI calculation logic.

Before generating the tests:

- Explain your testing approach.
- Identify the function(s) being tested.
- Describe why each test case is required.

Generate unit tests that cover:

- Valid inputs
- Invalid inputs
- Boundary values
- Empty inputs
- Decimal values

Use an appropriate JavaScript testing framework.

Do not modify the application code.

Task 5: Review AI-Generated Tests

Validate the generated tests before accepting them.

Suggested Prompt

Act as a Senior QA Engineer.

Review the generated unit tests.

Evaluate:

- Test coverage
- Missing business scenarios
- Missing edge cases
- Readability
- Maintainability
- Naming conventions
- Redundant tests
- Incorrect assumptions

Summarize your findings and recommend improvements.

Do not modify the generated tests.